EART11200: Natural Scientist's Toolkit

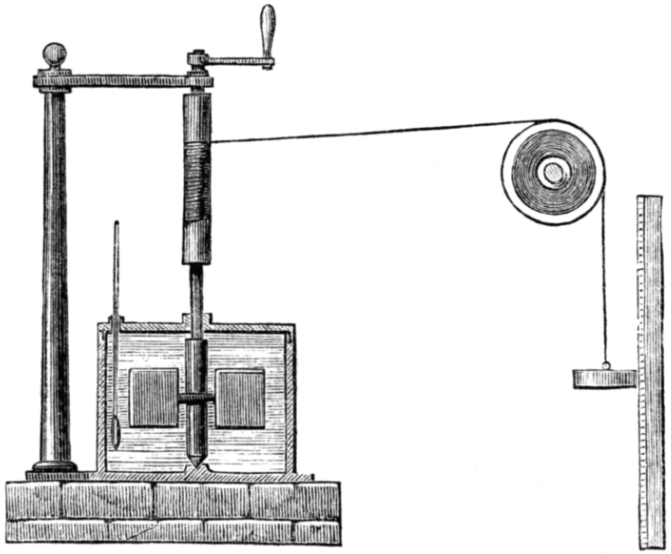


James Prescott Joule, 1818-1888

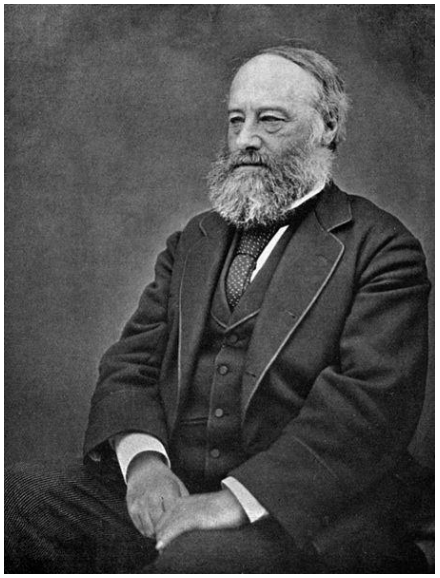
Before looking at the main problem for this week we need to consider the equivalence of heat and energy.

Heat and Energy equivalence

Paper in 1845 (***led to the law of conservation of energy***):



Joule's apparatus that demonstrated a falling weight converting potential energy, **=*mgh***, into viscous heating of water, **=mcδT**



James Prescott Joule, 1818-1888
Famous physicist and brewer, born in Salford.

Joule conducted many different experiments to demonstrate the quantitative equivalence of electrical, chemical and thermal phenomena, and that the forms of energy could be converted from one to the other.

'Believing that the power to destroy belongs to <u>The Creator</u> alone I affirm ... that any theory which, when carried out, demands the annihilation of force, is necessarily erroneous.'

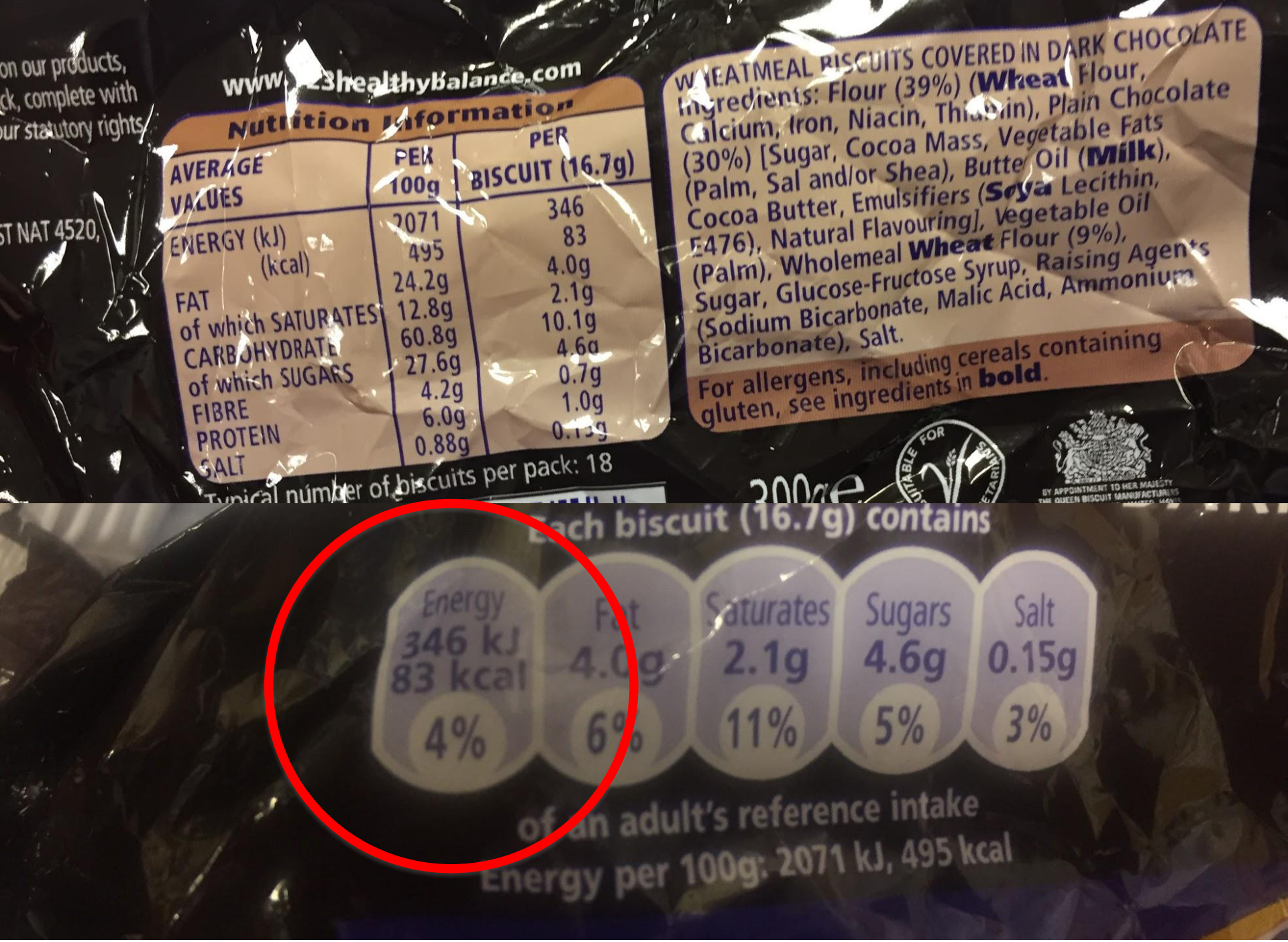
SI unit of energy is named the Joule, in recognition of this work.

Grown Biomass= Energy for power generation

Grown Biomass= Energy for food?






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ST NAT 4520,
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Nutrition Information
AVERAGE VALUES
PER 100g
PER BISCUIT (16.7g)
ENERGY (kJ) 2071 346
(kcal) 495 83
FAT 24.2g 4.0g
of which SATURATES 12.8g 2.1g
CARBOHYDRATE 60.8g 10.1g
of which SUGARS 27.6g 4.6g
FIBRE 4.2g 0.7g
PROTEIN 6.0g 1.0g
SALT 0.88g 0.15g
Typical number of biscuits per pack: 18
WHEATMEAL BISCUITS COVERED IN DARK CHOCOLATE
Ingredients: Flour (39%) (Wheat Flour, Calcium, Iron, Niacin, Thiamin), Plain Chocolate (30%) [Sugar, Cocoa Mass, Vegetable Fats (Palm, Sal and/or Shea), Butter Oil (Milk), Cocoa Butter, Emulsifiers (Soya Lecithin, E476), Natural Flavouring], Vegetable Oil (Palm), Wholemeal Wheat Flour (9%), Sugar, Glucose-Fructose Syrup, Raising Agents (Sodium Bicarbonate, Malic Acid, Ammonium Bicarbonate), Salt.
For allergens, including cereals containing gluten, see ingredients in bold.
300g
BY APPOINTMENT TO HER MAJESTY
Each biscuit (16.7g) contains
Energy 346 kJ 83 kcal 4%
Fat 4.0g 6%
Saturates 2.1g 11%
Sugars 4.6g 5%
Salt 0.15g 3%
of an adult's reference intake
Energy per 100g: 2071 kJ, 495 kcal

Energy is often quantified by its calorific content (e.g. Food).

- **calorie** (cal): amount of heat needed to raise the temperature of 1 gram of water 1 degree Celsius.
- **Calorie** (kcal): amount of heat needed to raise the temperature of 1 kilogram of water 1 degree Celsius.
- **calorie** = 4.18 joules of energy / heat
- This **heat content** can be used to fuel our activity, keep us warm, or used to make another form of energy (e.g. Electricity).

Energy required to raise the temperature of a substance

- Each substance has its own heat capacity

$$E = m \times C \times \Delta T$$

- The term ΔT is the change in temperature
- m is the mass
- C:specific heat capacity (=4180 J kg^{-1} K^{-1} for water)

Example – Joule's experiment

- Mass of 1 kg on string is 1.0 m above ground.
  - How much *potential energy* is there (answer in joules)?
  - If the mass then falls to the ground and drives paddles in water, mass 0.5 kg, what would the rise in temperature be? (note that heat capacity of water is 4180 joules per kg, per degC

2.2 In hot water

Problem 2.2 *While lying in the bath, you realise that not only are you paying for the energy to heat up the water but, since you have a water meter, you are paying directly for the water itself as well. Which of these is more expensive? Give estimates for heating the water either using electricity of gas.*

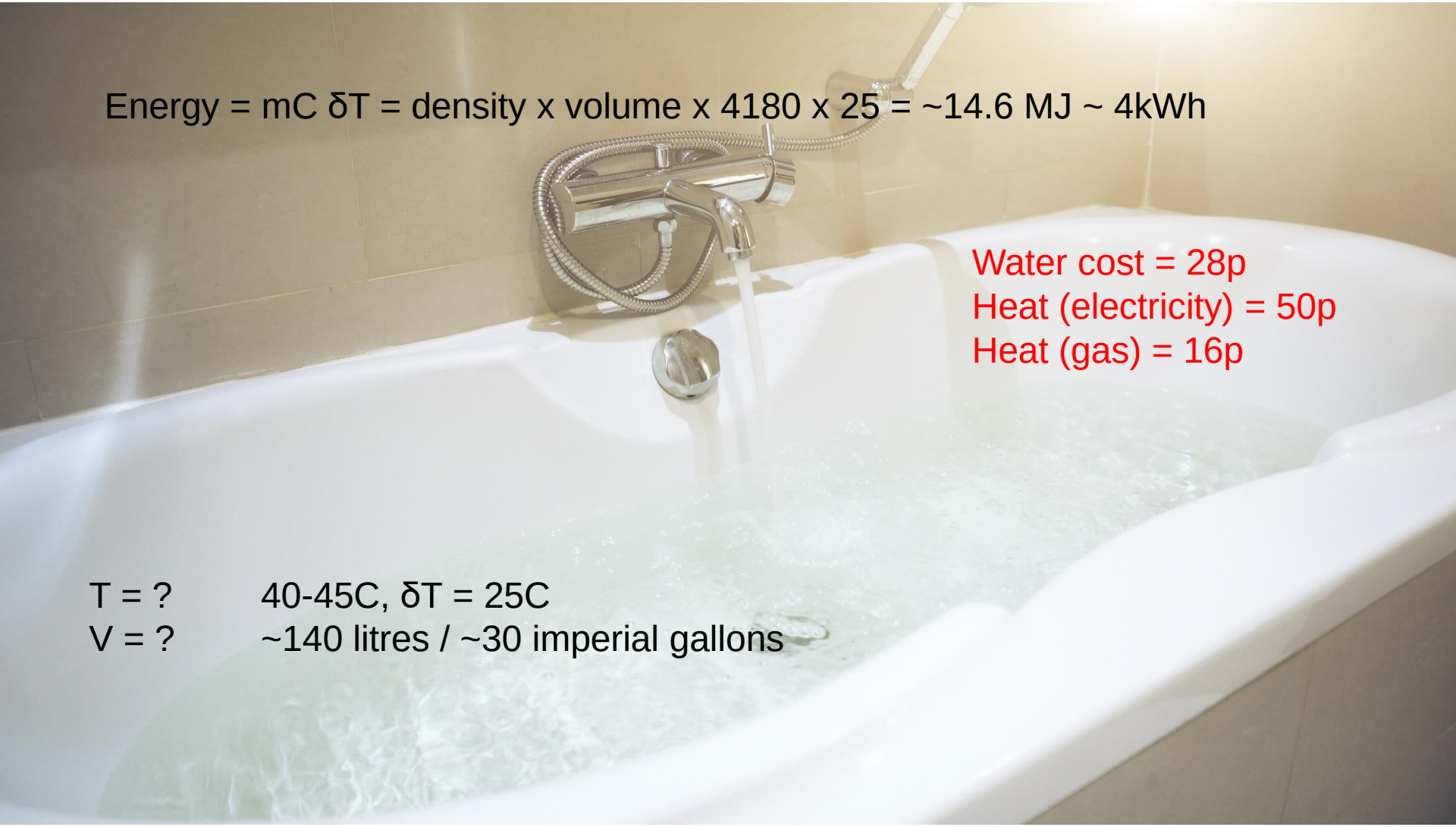
Assume that:

- The cost of water is £ 2 per m^{-3}.
- The cost of electricity per kWh is 12.5 pence.
- The cost of gas per kWh is 4 pence.
- Agree on temperature of water in the bath.
- Try to estimate the volume of water in the bath.

Things to consider:

- What would you guess is the cost of (a) the water and (b) the energy?
- What will you need to know to solve this problem? You should aim to get estimates of the costs of the two parts, so that you can compare them.
- Assume efficiency known (100% is OK) and not worrying about water left in the pipes, heat lost heating up the bathroom etc.

Unit conversions: 1kWh = 1000 x 60 x 60 = 3.6MJ; 1 m^3 water = 1000 l ~ 220 gallons
Constants: specific heat for water = 4180 $Jkg^{-1}K^{-1}$; density water ~ 1000 kgm^{-3}



Energy = mC δT = density x volume x 4180 x 25 = ~14.6 MJ ~ 4kWh

Water cost = 28p
Heat (electricity) = 50p
Heat (gas) = 16p

T = ? 40-45C, δT = 25C
V = ? ~140 litres / ~30 imperial gallons